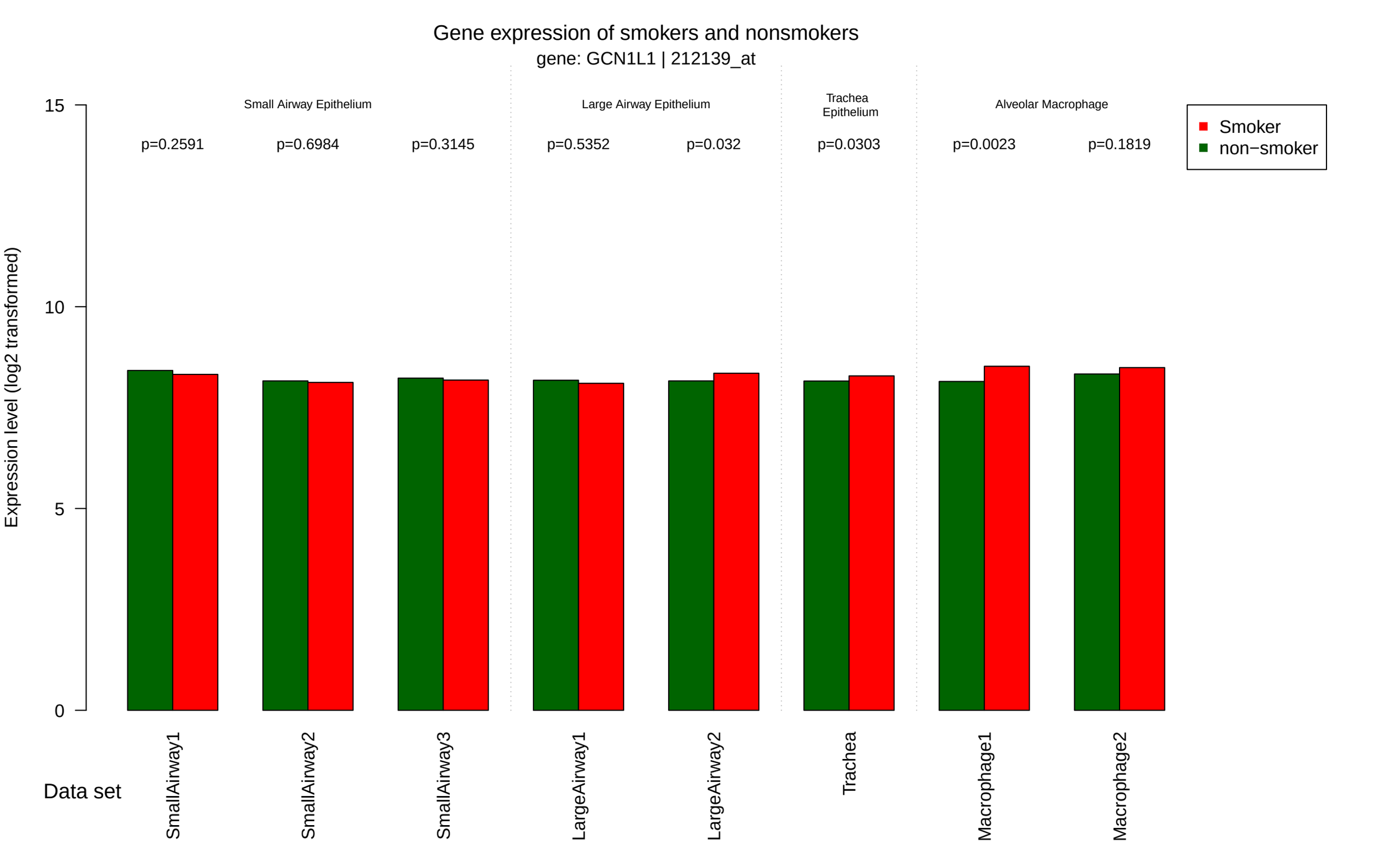
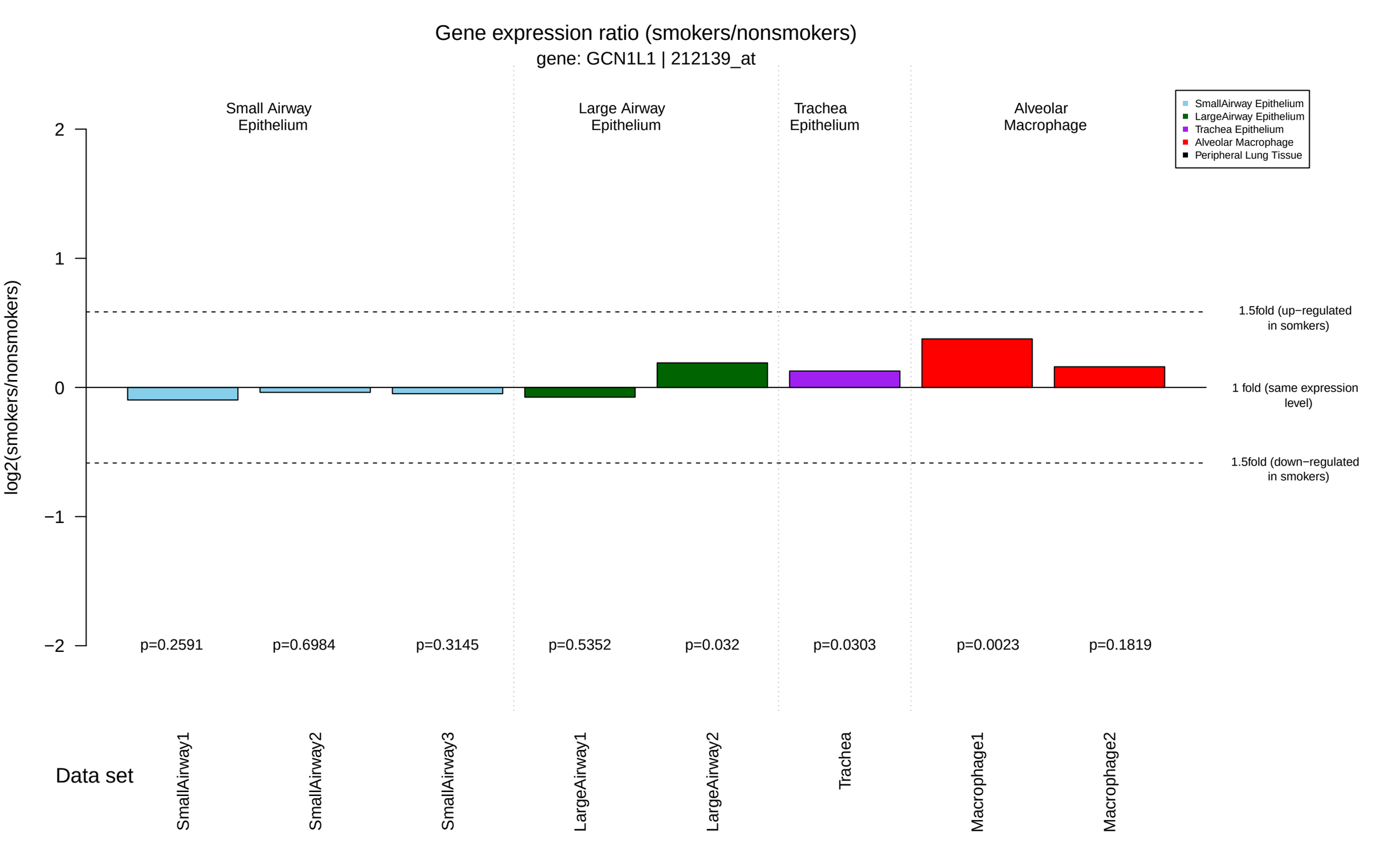
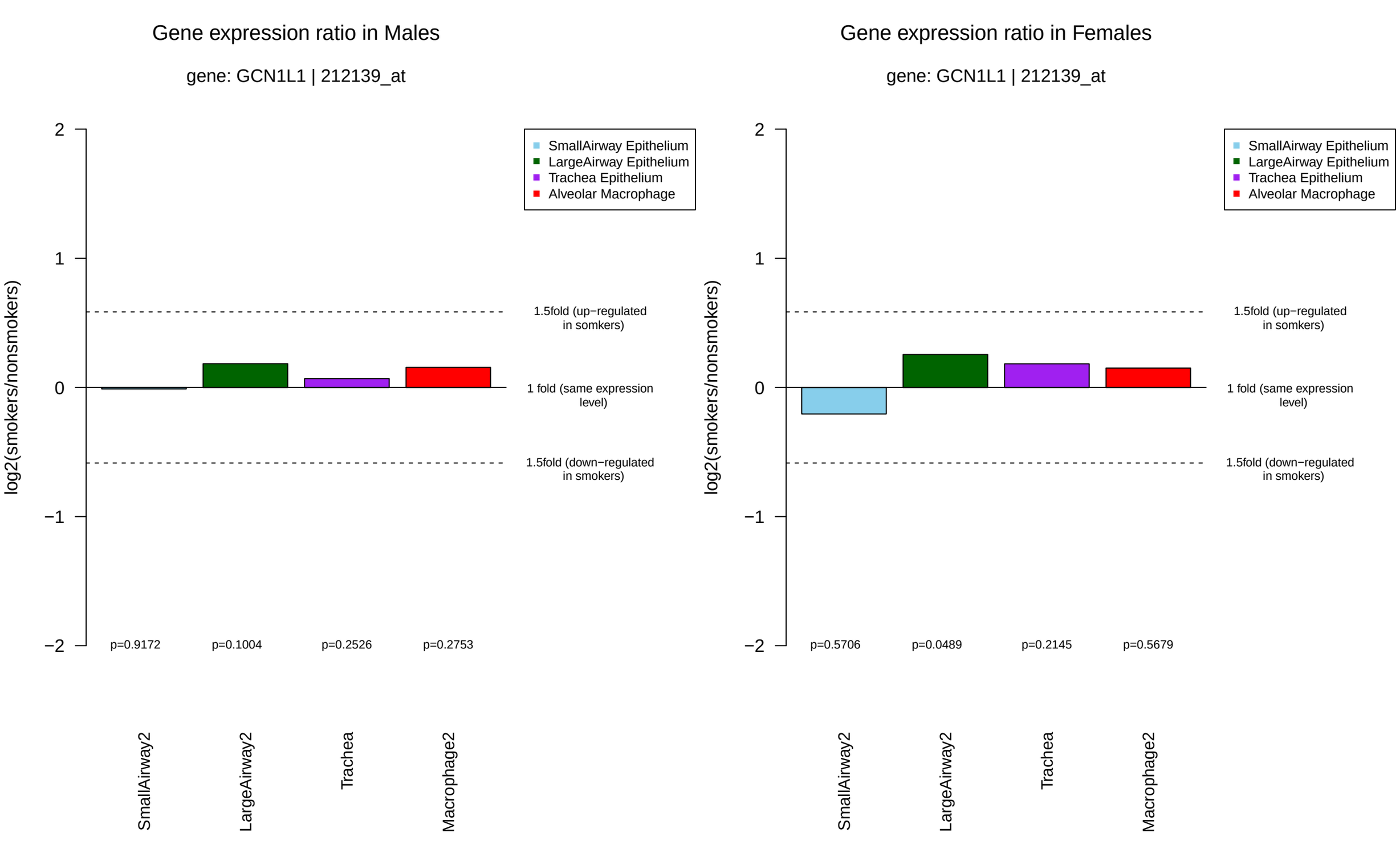
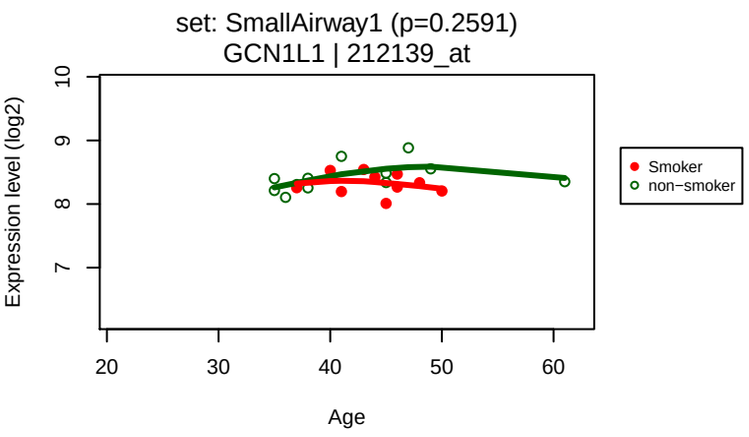
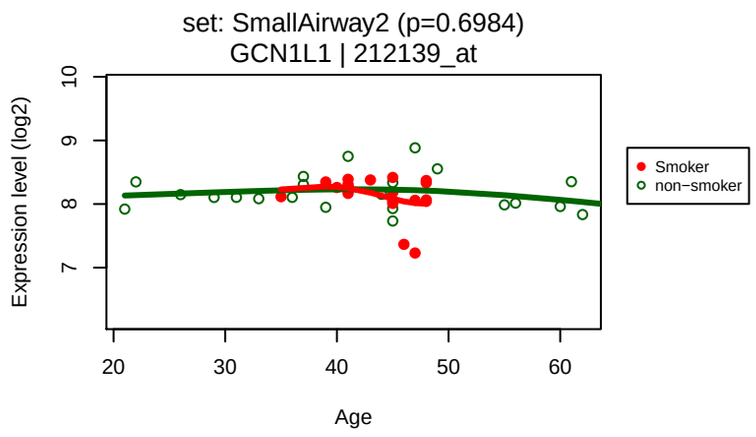
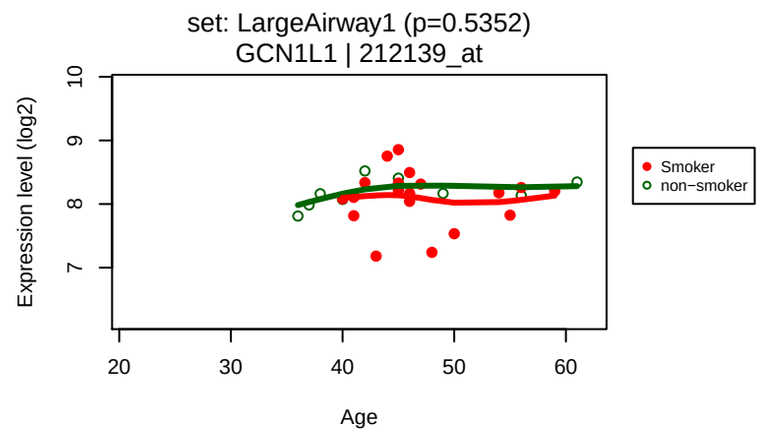

Gene expression of smokers and nonsmokers
gene: GCN1L1 | 212139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2591
p=0.6984
p=0.3145
p=0.5352
p=0.032
p=0.0303
p=0.0023
p=0.1819
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

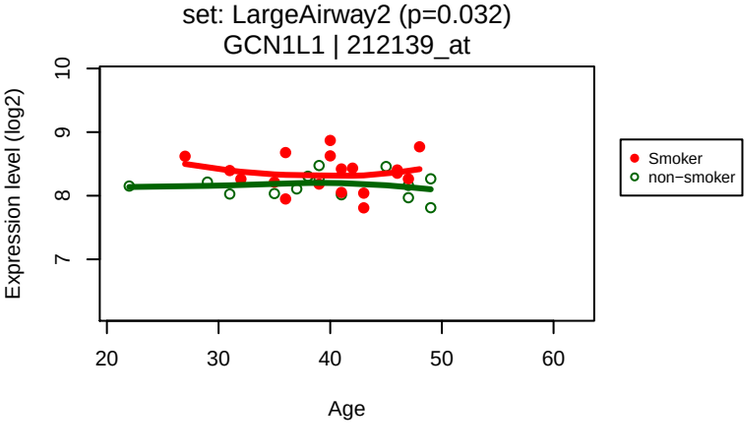

Gene expression ratio (smokers/nonsmokers)
gene: GCN1L1 | 212139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2591
p=0.6984
p=0.3145
p=0.5352
p=0.032
p=0.0303
p=0.0023
p=0.1819
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

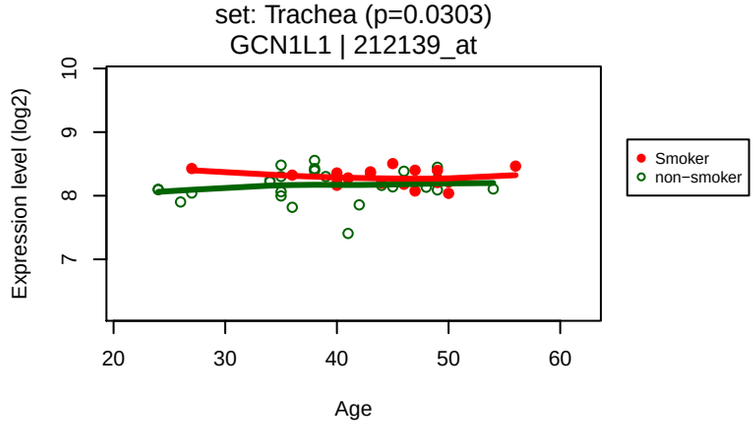

Gene expression ratio in Males
gene: GCN1L1 | 212139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9172
p=0.1004
p=0.2526
p=0.2753
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GCN1L1 | 212139_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5706
p=0.0489
p=0.2145
p=0.5679
SmallAirway2
LargeAirway2
Trachea
Macrophage2

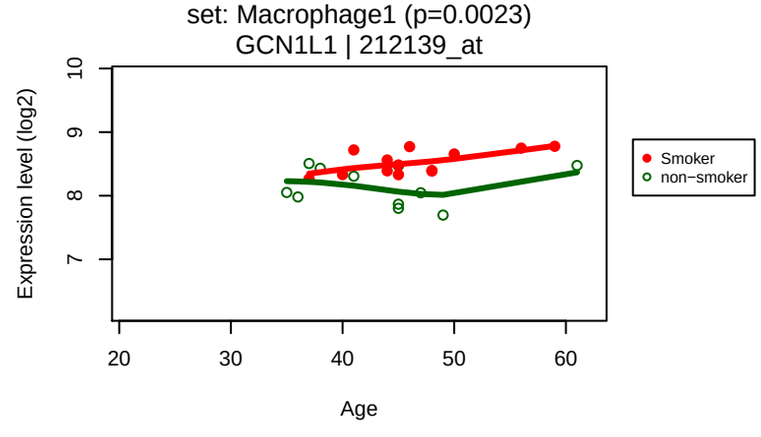

set: SmallAirway1 (p=0.2591)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker

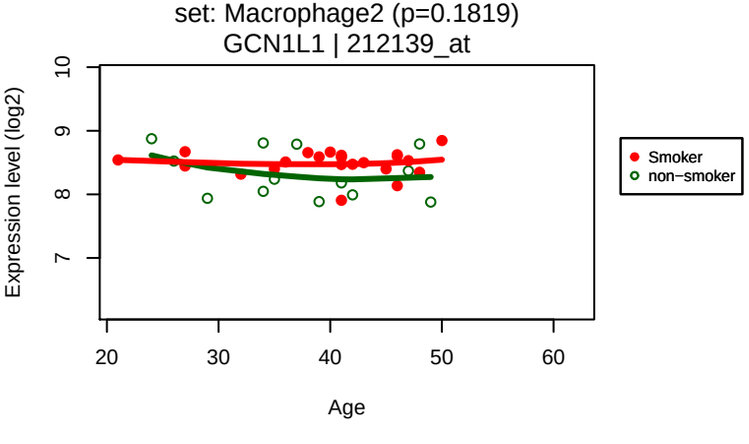

set: SmallAirway2 (p=0.6984)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker

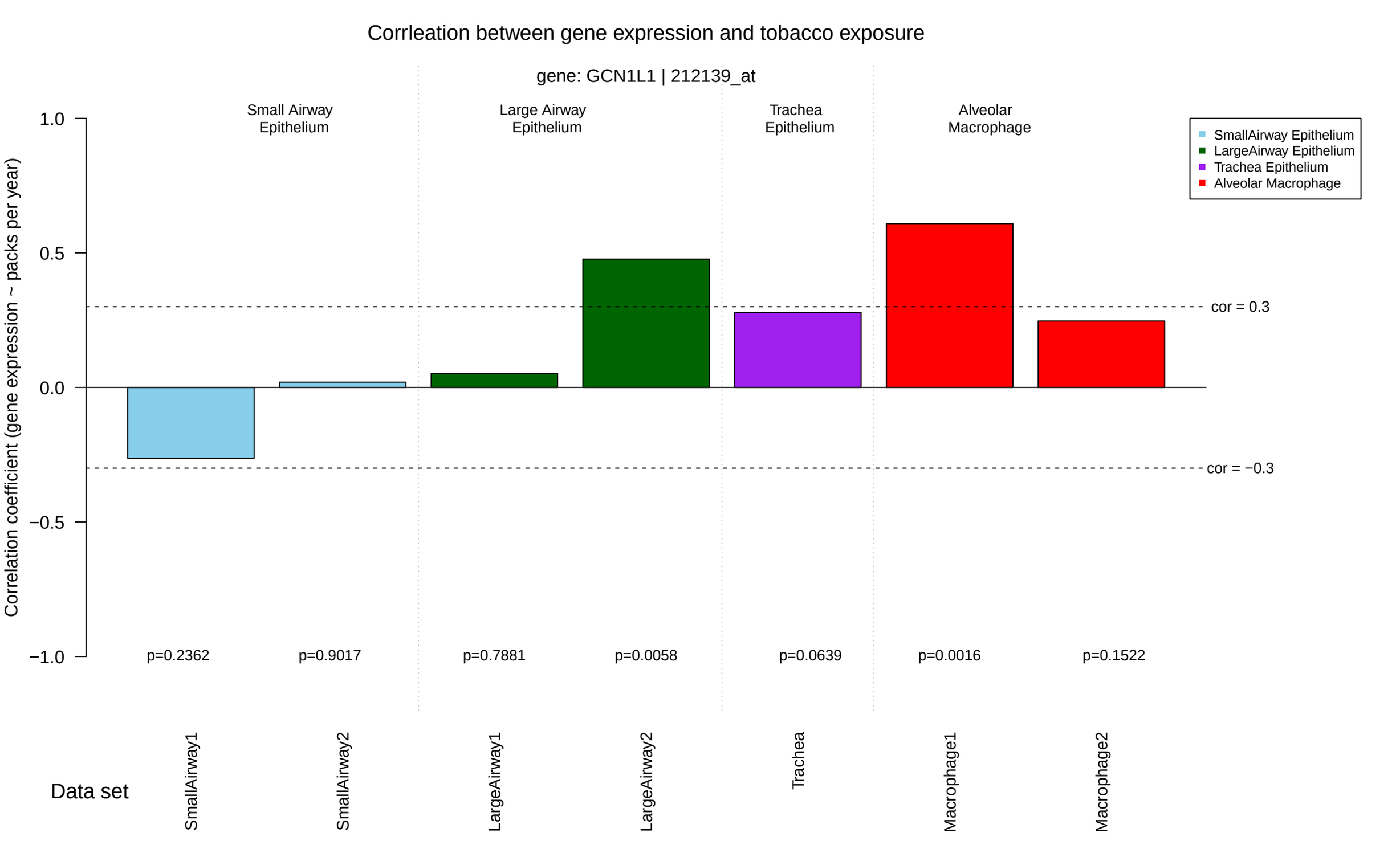

set: LargeAirway1 (p=0.5352)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.032)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0303)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0023)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1819)
GCN1L1 | 212139_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GCN1L1 | 212139_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2362
p=0.9017
p=0.7881
p=0.0058
p=0.0639
p=0.0016
p=0.1522
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2